

HTML5 Essential & Important Tags Study Guide

(High-Yield Reference for College Exams, Lab Viva & Assignments)

SECTION 1: DOCUMENT STRUCTURE & METADATA TAGS

1.1 `<html>`

- **Tag Type:** Paired / Container Tag (`<html>...</html>`)
- **Definition:** The root element that wraps all content in an HTML document.
- **Purpose:** Instructs browser that the document contains HTML code and defines language context.
- **Syntax & Example:**

```
<html lang="en">
  <!-- Document content -->
</html>
```

- **Output:** Defines document root boundary; no direct visual output.
- **Attributes:** `lang="en"` (Specifies primary document language).
- **Key Notes:** Must contain exactly one `<head>` and one `<body>` .

1.2 `<head>`

- **Tag Type:** Paired / Container Tag (`<head>...</head>`)
- **Definition:** Container for document metadata (data about the document).
- **Purpose:** Stores page title, character set, CSS stylesheets, scripts, and SEO tags.
- **Syntax & Example:**

```
<head>
  <meta charset="UTF-8">
  <title>College Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
```

- **Output:** Invisible on page body; configures browser settings and tab header.
- **Key Notes:** Executed/parsed before rendering the `<body>` .

1.3 `<title>`

- **Tag Type:** Paired Tag (`<title>...</title>`)
- **Definition:** Defines the title of the HTML document.
- **Purpose:** Displays text on browser tab, bookmark label, and search engine results title.
- **Syntax & Example:** `<title>Web Technologies Lab | CSE</title>`
- **Output:** Displays title string in browser top tab bar.
- **Key Notes:** Mandatory tag inside `<head>` . Exactly one title tag per page.

1.4 `<body>`

- **Tag Type:** Paired / Container Tag (`<body>...</body>`)

- **Definition:** Encapsulates all visible content of an HTML document.
- **Purpose:** Holds headings, paragraphs, images, tables, forms, and scripts shown to users.
- **Syntax & Example:**

```
<body>
  <h1>Welcome to CSE Portal</h1>
</body>
```

- **Output:** Renders full visual canvas of the web page.

1.5 <meta>

- **Tag Type:** Void / Self-Closing Tag (<meta>)
- **Definition:** Provides metadata about character encoding, page description, and viewport layout.
- **Purpose:** Ensures UTF-8 symbol support and responsive layout on mobile screens.
- **Syntax & Examples:**

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- **Key Attributes:**
 - `charset="UTF-8"` : Unicode encoding.
 - `name="viewport"` / `content="width=device-width, initial-scale=1.0"` : Responsive mobile rendering.

1.6 <link>

- **Tag Type:** Void / Self-Closing Tag (<link>)
- **Definition:** Links external resources to current HTML document.
- **Purpose:** Attaches external CSS stylesheets or website favicons.
- **Syntax & Example:** `<link rel="stylesheet" href="styles.css">`
- **Key Attributes:** `rel="stylesheet"` , `href="URL"` , `type="text/css"` .

SECTION 2: HEADINGS, TEXT & FORMATTING TAGS

2.1 <h1> to <h6>

- **Tag Type:** Paired Tags (<h1>...</h1> to <h6>...</h6>)
- **Definition:** Six levels of document headings (<h1> highest priority, <h6> lowest).
- **Purpose:** Organizes content hierarchy and signals topic importance to search engines.
- **Syntax & Example:**

```
<h1>Module 1: HTML5 (Main Heading)</h1>
<h2>1.1 Introduction (Sub-heading)</h2>
```

- **Output:** Bold text block rendered in decreasing font sizes from H1 to H6.

2.2 <p> (Paragraph)

- **Tag Type:** Paired Tag (<p>...</p>)
- **Definition:** Represents a paragraph block of text.

- **Purpose:** Groups related text sentences with standard top and bottom vertical margin spacing.
- **Example:** `<p>HTML is the standard markup language for Web pages.</p>`

2.3 `<br>` & `<hr>`

- **Tag Type:** Void Tags (`<br>` , `<hr>`)
- **Definitions:**
 - `<br>` : Inserts a single text line break.
 - `<hr>` : Inserts a horizontal rule line break (thematic divider).
- **Example:**

```
<p>Line 1<br>Line 2</p>
<hr>
```

2.4 `<a>` (Anchor / Hyperlink)

- **Tag Type:** Paired Tag (`<a>...</a>`)
- **Definition:** Creates a clickable hyperlink to navigate to another page, file, or section.
- **Purpose:** Enables web navigation.
- **Syntax & Example:** `<a href="https://vtu.ac.in" target="_blank">Visit VTU Website</a>`
- **Key Attributes:**
 - `href="URL"` : Target location link destination.
 - `target="_blank"` : Opens link in a new browser tab.
 - `download` : Prompts file download.

2.5 `<img>` (Image)

- **Tag Type:** Void Tag (`<img>`)
- **Definition:** Embeds an image file into the webpage layout.
- **Syntax & Example:** ``
- **Key Attributes:**
 - `src="path"` : Image file source URL (Mandatory).
 - `alt="text"` : Alternate text description for accessibility & broken images (Mandatory).
 - `width` / `height` : Image dimensions in pixels.

2.6 `<strong>` & `<em>` (Semantic Formatting)

- **Tag Type:** Paired Inline Tags (`<strong>...</strong>` , `<em>...</em>`)
- **Definitions:**
 - `<strong>` : Indicates strong importance (Renders as **bold**).
 - `<em>` : Indicates vocal stress emphasis (Renders as *italics*).
- **Difference from `<b>` & `<i>`** : `<strong>` and `<em>` carry semantic meaning for screen readers, whereas `<b>` and `<i>` are purely visual styling.

2.7 `<pre>` & `<code>`

- **Tag Type:** Paired Tags (`<pre>...</pre>` , `<code>...</code>`)
- **Definitions:**
 - `<pre>` : Preserves exact white spaces, tabs, and line breaks.
 - `<code>` : Formats inline code snippet in monospace font.
- **Combined Example:**

```
<pre><code>
#include <istream>
int main() { return 0; }
</code></pre>
```

SECTION 3: LIST TAGS

3.1 Unordered (`<ul>`) & Ordered (`<ol>`) Lists

- **Tag Type:** Container Tags (`<ul>` , `<ol>`) paired with `<li>` (List Item).
- **Definitions:**
 - `<ul>` : Bulleted list where order does not matter.
 - `<ol>` : Numbered sequential list.
- **Examples:**

```
<!-- Unordered List -->
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<!-- Ordered List -->
<ol type="1" start="1">
  <li>Step 1</li>
  <li>Step 2</li>
</ol>
```

- **Key Attributes in `<ol>` :** `type="1|A|a|I|i"` , `start="number"` , `reversed` .

3.2 Description List (`<dl>` , `<dt>` , `<dd>`) (Frequent Exam Question)

- **Tag Type:** Paired Container Tags.
- **Definition:** Renders a list of terms (`<dt>`) and their matching descriptions (`<dd>`).
- **Example:**

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

SECTION 4: TABLE TAGS (LAB VIVA ESSENTIAL)

4.1 `<table>`, `<tr>`, `<th>`, `<td>`, `<caption>`

- **Definitions:**

- `<table>` : Container grid for tabular data.
- `<caption>` : Table heading title caption.
- `<tr>` : Table Row.
- `<th>` : Table Header Cell (Bold & centered).
- `<td>` : Table Data Cell.

- **Example with `colspan` & `rowspan` :**

```
<table border="1">
  <caption>Student Marks Sheet</caption>
  <tr>
    <th>USN</th>
    <th>Name</th>
    <th colspan="2">Marks (Lab + Theory)</th>
  </tr>
  <tr>
    <td>1VU22CS001</td>
    <td>Aryansh</td>
    <td>48</td>
    <td>95</td>
  </tr>
</table>
```

- **Key Cell Attributes:**

- `colspan="n"` : Merges n adjacent columns horizontally.
- `rowspan="n"` : Merges n adjacent rows vertically.

SECTION 5: FORMS & INPUT CONTROLS (HIGHEST WEIGHTAGE)

5.1 `<form>`

- **Tag Type:** Paired Container Tag (`<form>...</form>`)
- **Definition:** Container used to collect user input data and transmit it to a web server.
- **Syntax & Example:**

```
<form action="/submit-data.php" method="POST">
  <!-- Input controls -->
</form>
```

- **Key Attributes:**

- `action="URL"` : Target server script endpoint.
- `method="GET|POST"` : HTTP method. (`GET` appends data in URL; `POST` sends data securely in HTTP request payload body).
- `enctype="multipart/form-data"` : Mandatory when uploading files via `<input type="file">`.

5.2 `<input>` Types & Attributes

- **Tag Type:** Void Tag (`<input>`)

• Essential Input Types Table:

type Value	Purpose	Example
text	Single-line plain text entry.	<code><input type="text" name="username"></code>
password	Masks text input with dots/asterisks.	<code><input type="password" name="pass"></code>
email	Email address entry with auto-validation.	<code><input type="email" name="useremail"></code>
number	Numeric input with min/max bounds.	<code><input type="number" min="1" max="100"></code>
radio	Radio button (select ONLY ONE from group).	<code><input type="radio" name="gender" value="male"></code>
checkbox	Checkbox toggle (select MULTIPLE choices).	<code><input type="checkbox" name="skills" value="html"></code>
file	File attachment picker.	<code><input type="file" name="resume"></code>
submit	Submits form to server.	<code><input type="submit" value="Submit"></code>
reset	Resets form fields to default values.	<code><input type="reset" value="Clear"></code>

• Essential Input Attributes:

- `name="str"` : Variable key sent to server payload (Mandatory).
- `value="str"` : Initial or submitted value.
- `placeholder="text"` : Hint text shown inside empty input box.
- `required` : Prevents form submission if field is blank.
- `readonly` : Field is readable but user cannot edit text.
- `disabled` : Deactivates field and excludes it from form submission payload.

5.3 <label>

- **Tag Type:** Paired Tag (`<label>...</label>`)
- **Purpose:** Binds readable text label to an input field for accessibility.
- **Example:**

```
<label for="student-usn">Enter USN:</label>
<input type="text" id="student-usn" name="usn">
```

- **Key Attribute:** `for="id_value"` (Must match target input's `id` attribute).

5.4 <select> & <option> (Dropdown Menu)

- **Example:**

```
<label for="branch">Select Branch:</label>
<select id="branch" name="branch">
  <option value="cse">Computer Science</option>
  <option value="ise">Information Science</option>
</select>
```

5.5 `<textarea>` (Multi-line Text Area)

- **Example:** `<textarea name="address" rows="4" cols="30" placeholder="Enter full address...">`
`</textarea>`

SECTION 6: SEMANTIC LAYOUT TAGS (MODERN HTML5)

HTML5 introduced semantic tags to give clear meaning to web layout sections instead of using generic `<div>` elements for everything.

Tag	Purpose & Description
<code><header></code>	Represents top header banner containing title, logo, and main navigation.
<code><nav></code>	Defines container for primary navigation hyperlink menu bar.
<code><main></code>	Encapsulates the unique primary content of the web page (Max 1 per page).
<code><section></code>	Groups related content logically into thematic chapters or topics.
<code><article></code>	Represents a self-contained, independent post or article.
<code><aside></code>	Defines sidebar or callout box with related peripheral content.
<code><footer></code>	Defines bottom footer area containing copyright, terms, and contact details.
<code><div></code>	Generic non-semantic block-level container used for CSS styling.
<code></code>	Generic non-semantic inline container used for inline text styling.

SECTION 7: MULTIMEDIA & INTERACTIVE TAGS

7.1 `<video>` & `<audio>`

- **Example:**

```
<!-- Video Player -->
<video width="320" height="240" controls poster="thumb.jpg">
  <source src="lecture.mp4" type="video/mp4">
  Your browser does not support video playback.
</video>
```

- **Key Attributes:** `controls` (shows play/pause buttons), `autoplay`, `loop`, `muted`.

7.2 `<iframe>` (Inline Frame)

- **Definition:** Embeds an external web page or YouTube video window inside current page.
- **Example:** `<iframe src="https://maps.google.com" width="400" height="300" title="Map"></iframe>`

7.3 `<details>` & `<summary>` (Accordion Widget)

- **Example:**

```
<details>
  <summary>Click to view Exam Rule</summary>
  <p>Bring college ID card to examination hall.</p>
</details>
```

SECTION 8: QUICK REVISION & COMPARISON SUMMARY

1. Block-Level vs Inline Elements

Feature	Block-Level Elements	Inline Elements
Line Break	Starts on a new line ; forces newline after.	Fits inline with surrounding text.
Width	Takes 100% full container width .	Takes ONLY width required by content.
Examples	<code><div></code> , <code><p></code> , <code><h1>-<h6></code> , <code></code> , <code><table></code> , <code><form></code>	<code></code> , <code><a></code> , <code></code> , <code></code> , <code></code> , <code><input></code>

2. Common Void (Self-Closing) Elements List

Void elements **cannot contain child text/content** and **do NOT have closing tags**:

- `<br>`, `<hr>`, `<img>`, `<input>`, `<link>`, `<meta>`, `<source>`

SECTION 9: COMPLETE EXAM PRACTICE HTML PAGE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration Portal</title>
</head>
<body>

  <header>
    <h1>Department of Computer Science</h1>
    <nav>
      <a href="#about">About</a> | <a href="#register">Register</a>
    </nav>
  </header>

  <main>
    <section id="about">
      <h2>Web Technology Lab Course</h2>
      <p>Learn <strong>HTML5</strong>, <em>CSS3</em>, and JavaScript.</p>
    </section>

    <section id="register">
      <h2>Student Registration Form</h2>
      <form action="/submit" method="POST">
        <p>
          <label for="usn">USN:</label><br>
          <input type="text" id="usn" name="usn" required placeholder="e.g. 1VU22CS001">
        </p>
        <p>
          <label for="branch">Branch:</label><br>
          <select id="branch" name="branch">
            <option value="cse">CSE</option>
            <option value="ise">ISE</option>
          </select>
        </p>
        <p>
          <input type="submit" value="Register Student">
        </p>
      </form>
    </section>
  </main>

  <footer>
    <p>&copy; 2026 College Web Portal. All rights reserved.</p>
  </footer>

</body>
</html>
```